Arc length

1

a Arc
b Sector
c Chord
d Arc
e Semicircle

2

a
i $\frac{1}{2}$
ii 9π cm

b
i $\frac{1}{4}$
ii 4π cm

c
i $\frac{1}{3}$
ii 6π cm

d
i $\frac{1}{6}$
ii 2π cm

3 64 cm

4 20 cm

5 $\frac{3799\pi}{180}$ m

Area and perimeter of a sector

6

a No
b No
c Yes
d No
e Yes
f Yes

7

a $\frac{8}{45}$
b $\frac{71}{180}$
c $\frac{143}{180}$
d $\frac{27}{40}$
e $\frac{41}{360}$
f $\frac{1}{4}$

8

a 25.13 cm^2
b 12.57 cm^2

9

a
i $\frac{1}{6}$
ii $6\pi\text{ cm}^2$

b
i $\frac{1}{12}$
ii $3\pi\text{ cm}^2$

10

11

a

i $\frac{1}{3}$ ii $85.98\,\text{cm}$ iii $461.81\,\text{cm}^2$

b

i $\frac{1}{6}$ ii $73.13\,\text{cm}$ iii $301.59\,\text{cm}^2$

12

a $\frac{1}{4}$ b 7π cm c 49.99 cm

13 $14\pi + 28\,\text{cm}$

14

a

i 7 cm ii $3\ \text{cm}^2$

b

i 27 cm ii $38\ \text{cm}^2$

c

i 12 cm ii $9\ \text{cm}^2$

d

i 37 cm ii $75\ \text{cm}^2$

e

i 41 cm ii $101\ \text{cm}^2$

f

i 26 cm ii $39\ \text{cm}^2$

15

a 10.28 cm b 97.69 cm c 20.57 cm d 44.29 cm

e 17.85 cm f 60.70 cm g 281.92 cm h 36.85 cm

i 11.22 cm j 117.59 cm

Annulus

16

a No b No c No d Yes

e Yes f Yes

17

a

i 6 cm ii 10 cm

b

i 4 cm ii 7.5 cm

18

a 8 cm b 12 cm c 75.4 cm d 50.3 cm e $125.7\,\text{cm}$

f $251.3\,\text{cm}^2$

19

a $39\pi\,\text{cm}^2$

b $122.52\,\text{cm}^2$

20

a $113.1\ \text{cm}^2$

b $40.8\ \text{cm}^2$

21 $38\pi\,\text{cm}$

22 $91.11\,\text{cm}$

23 $56\pi\,\text{cm}^2$

24

a $125.7\ \text{cm}^2$

b Brad is incorrect, because the new logo's area can be found in the same way as the first logo. By finding the area of the outer circle and subtracting the area of the inner circle. So it has the same area.

c Homer is correct, the new logo is not an annulus because the width of the grey area is not the same all the way around.